RowHammer

Peter Deutsch

Spring 2026

RowHammer In One Sentence

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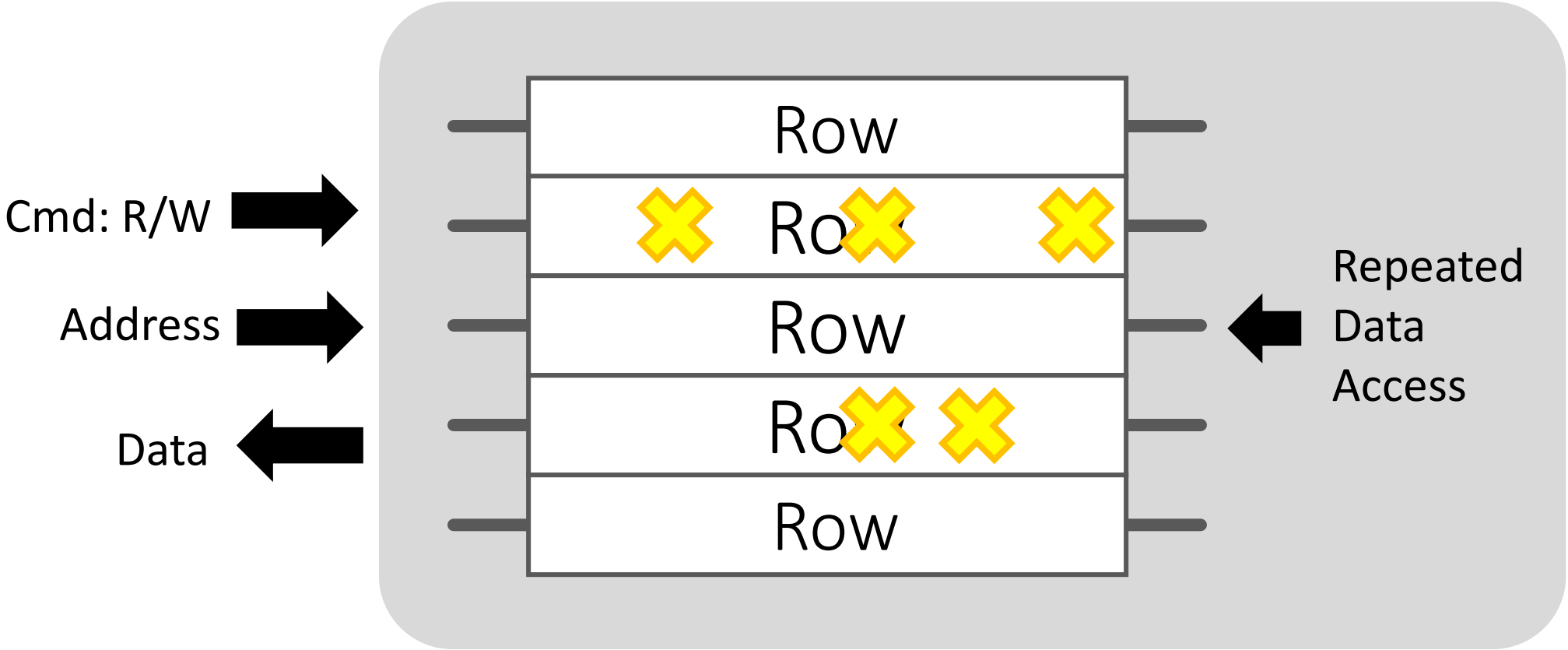

Observation: Repeatedly accessing a row enough times can cause disturbance errors in nearby rows

Outline

- Why does RowHammer happen? What is its working mechanism?
- How to trigger RowHammer in practice? Any challenges? -> Lab 4
- If you are an attacker, how can you exploit it?
- If you are a defender, what can you do? -> Next Lecture

DRAM Basics

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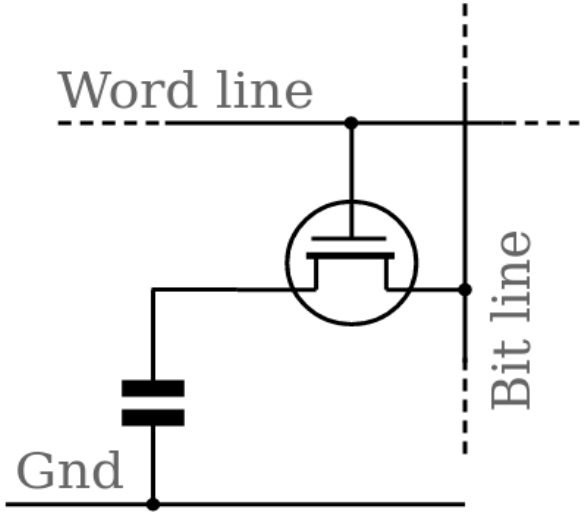
- Each bit in DRAM is stored in a “cell” using a *capacitor*
- *Read is **destructive***
- DRAM cells lose their state over time (hence ***Dynamic*** RAM)
- Data stored in DRAM cells needs to be “**refreshed**” at a regular interval



https://www.electronics-notes.com/articles/electronic_components/semiconductor-ic-memory/dynamic-ram-how-does-dram-work-operation.php

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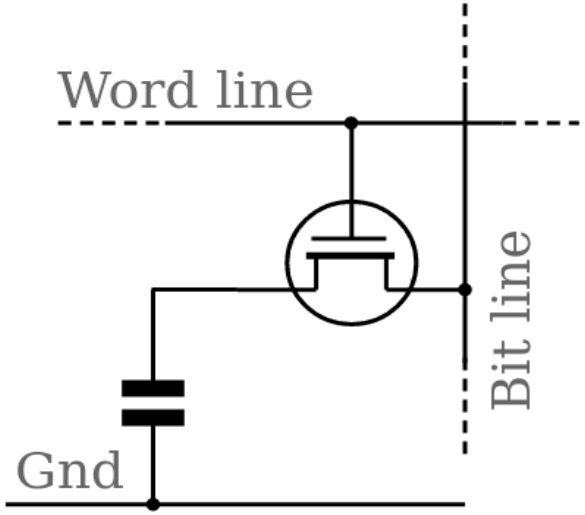
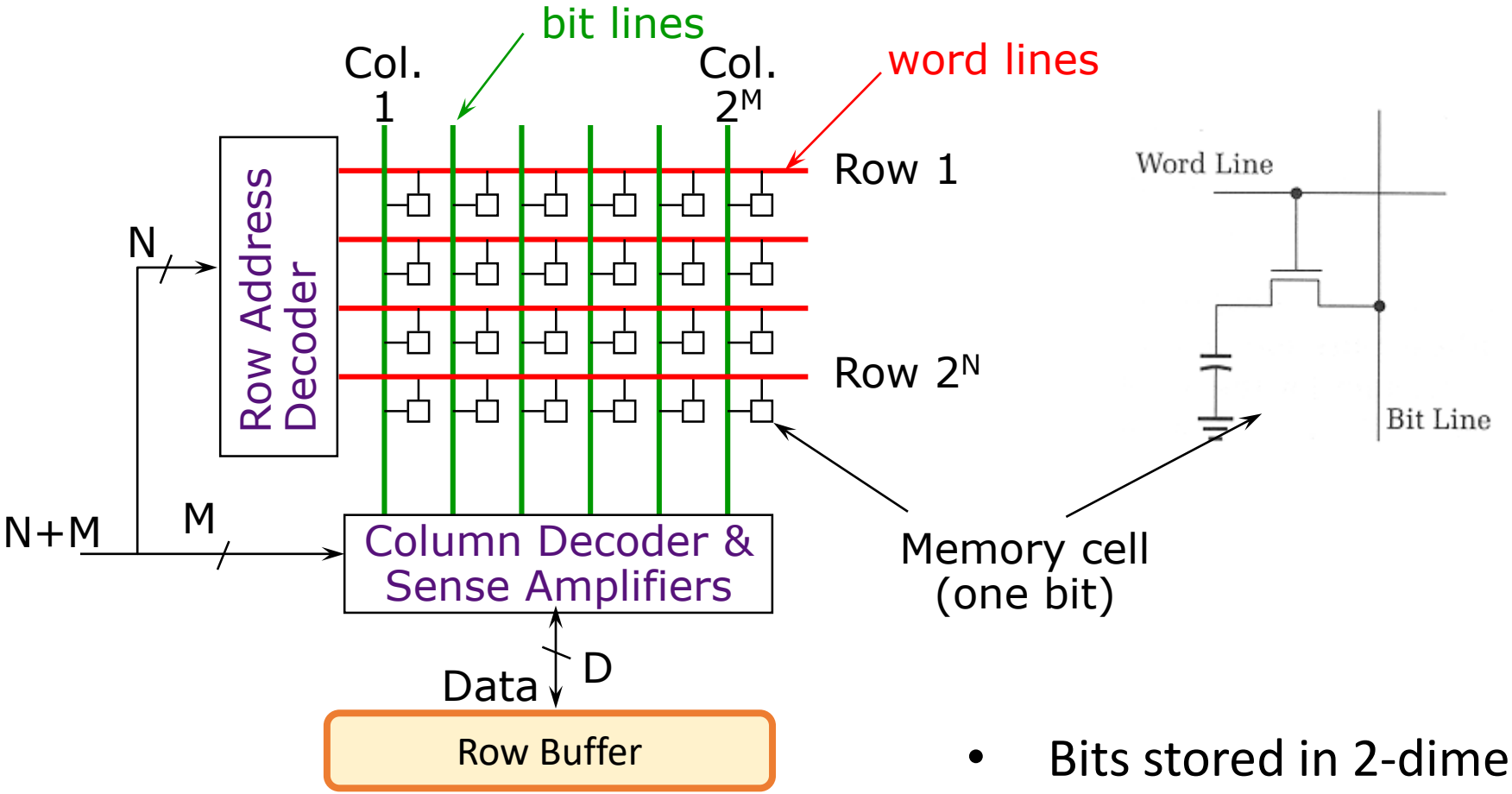


Why we widely use DRAM given some of its unappealing properties? (compare it with SRAM and SSD)

- Speed (2-10x slower than SRAM)
- Density (20x denser than SRAM)
- Cost (~100x cheaper per MB)

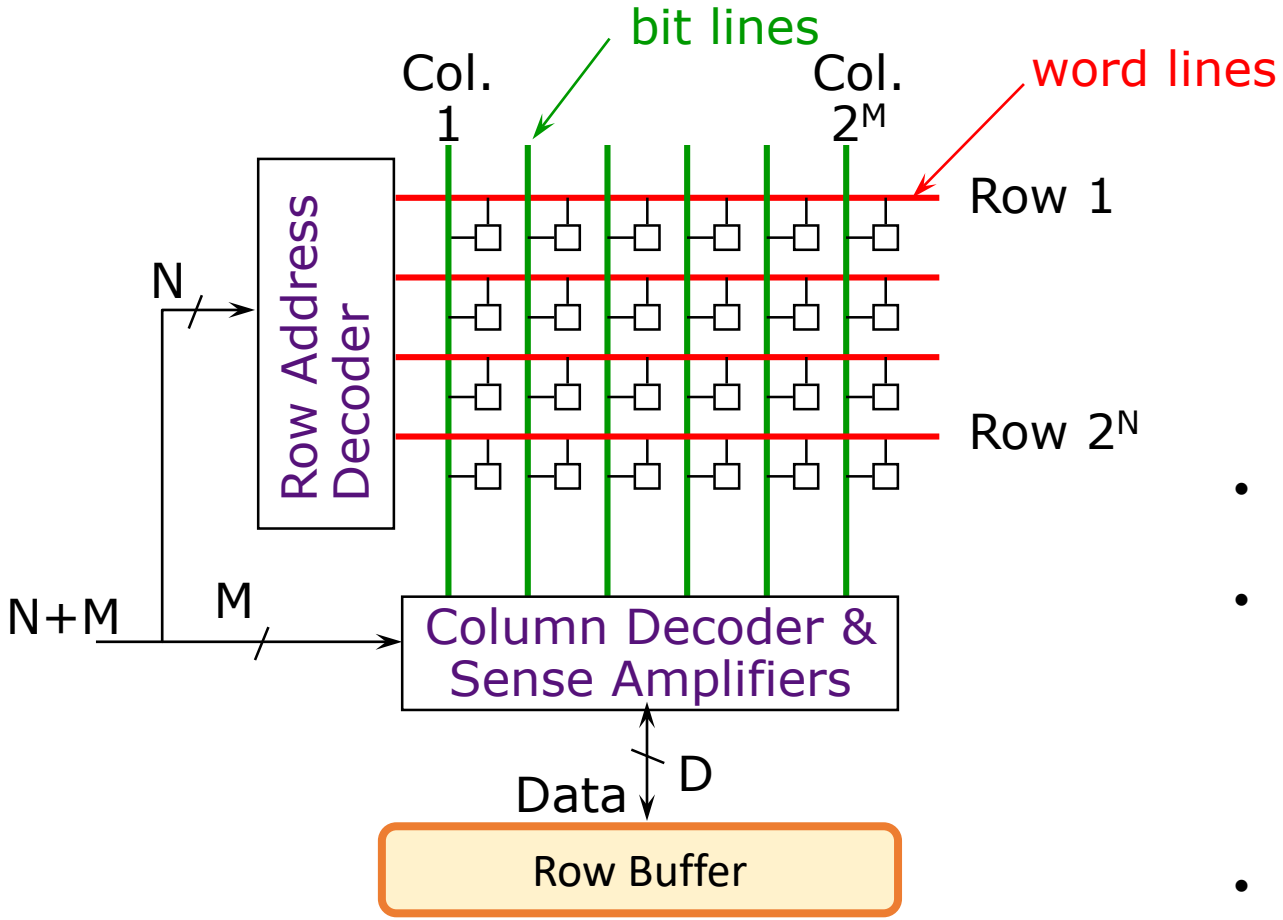
https://www.electronics-notes.com/articles/electronic_components/semiconductor-ic-memory/dynamic-ram-how-does-dram-work-operation.php

DRAM Architecture



- Bits stored in 2-dimensional arrays on chip
- Question: why read the entire row?

DRAM Refresh



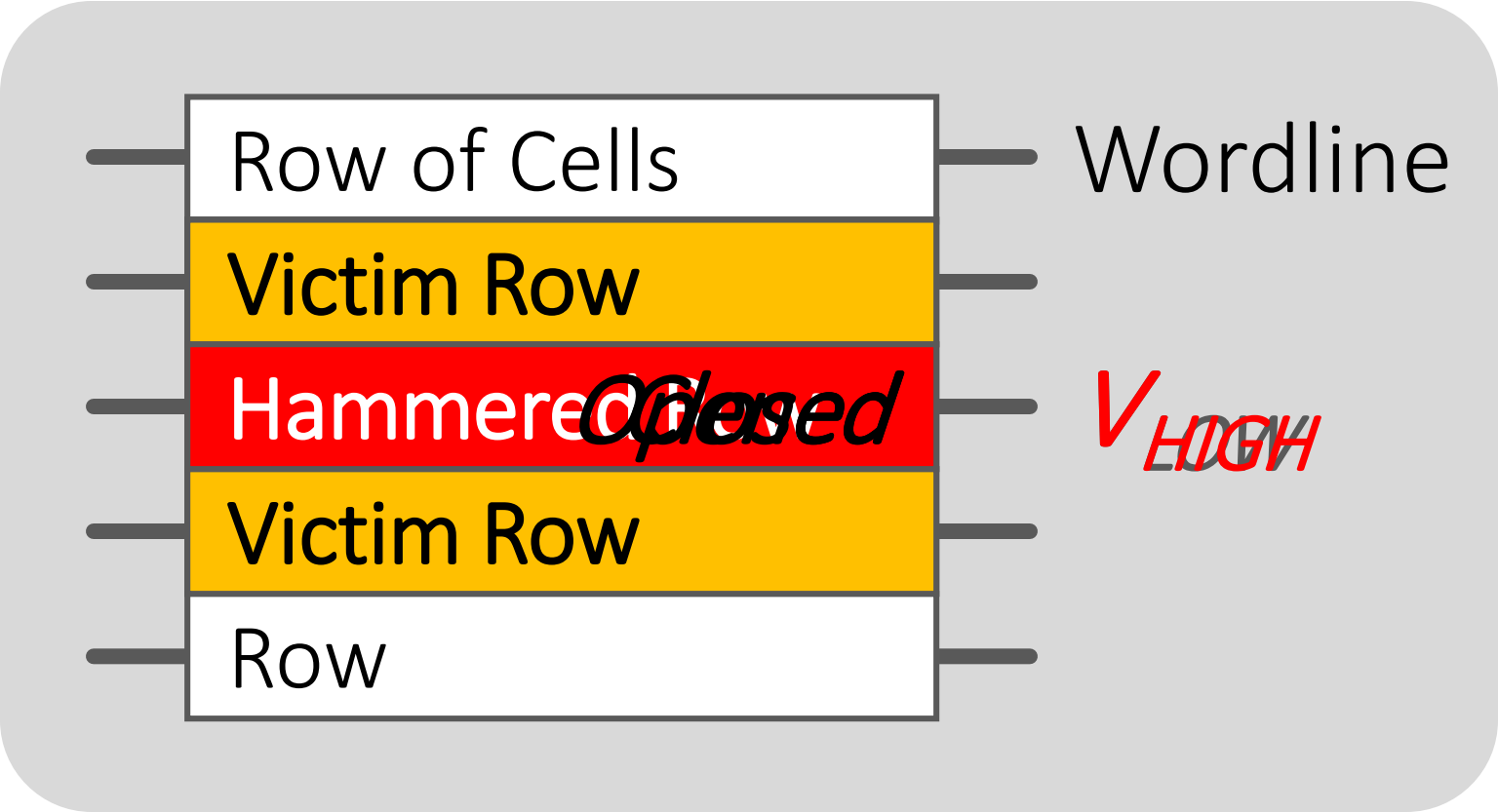
- How to do refresh?
- Performance penalty of refresh
  - In an 8Gb memory, upwards of 10% of time is spent in refresh!
- The common refresh interval: **64ms**

Aside: Cold Boot Attacks

	Seconds w/o power	Error % at operating temp.	Error % at −50 °C
SDRAM (1999)	60	41	(no errors)
	300	50	0.000095
DDR (2001)	360	50	(no errors)
	600	50	0.000036
DDR (2003)	120	41	0.00105
	360	42	0.00144
DDR2 (2007)	40	50	0.025
	80	50	0.18

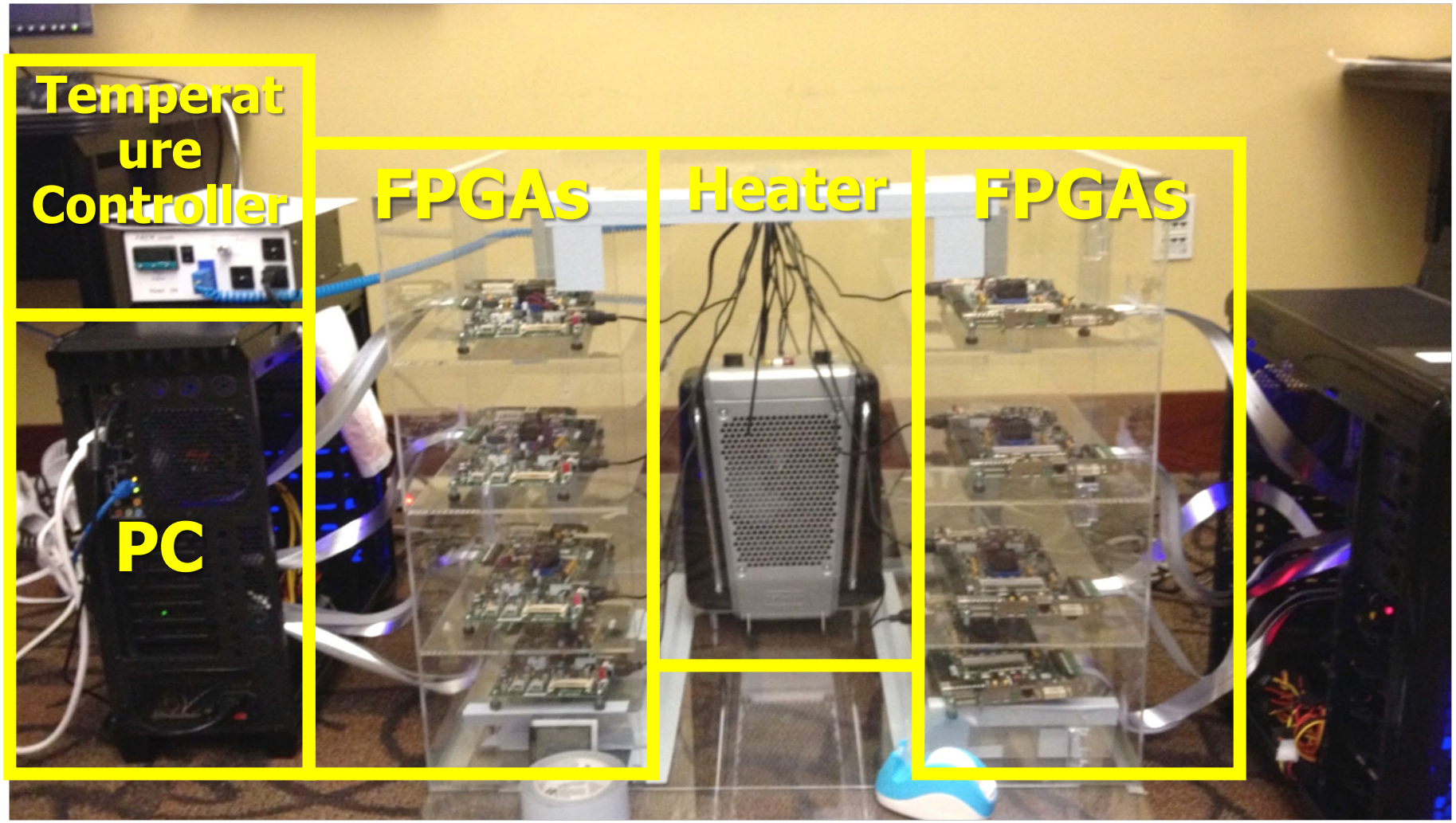
Halderman et al.; Lest We Remember: Cold Boot Attacks on Encryption Keys; USENIX Security'08

See RowHammer Again



Observation: Repeatedly accessing a row enough times **between refreshes** can cause disturbance errors in nearby rows

Infrastructures to Understand Rowhammer



Kim et al; Flipping Bits in Memory Without Accessing Them: An Experimental Study of DRAM Disturbance Errors; ISCA'14

Slide credit to Onur Mutlu

Most DRAM Modules Are Vulnerable

A company	B company	C company
86% (37/43)	83% (45/54)	88% (28/32)
Up to 1.0×10^{7} errors	Up to 2.7×10^{6} errors	Up to 3.3×10^{5} errors

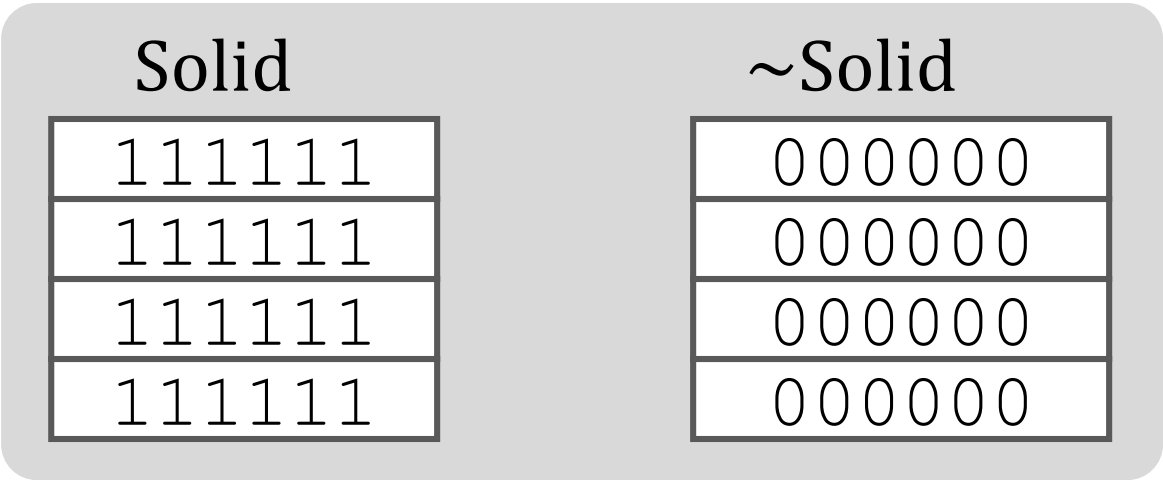
Kim et al; Flipping Bits in Memory Without Accessing Them: An Experimental Study of DRAM Disturbance Errors; ISCA'14

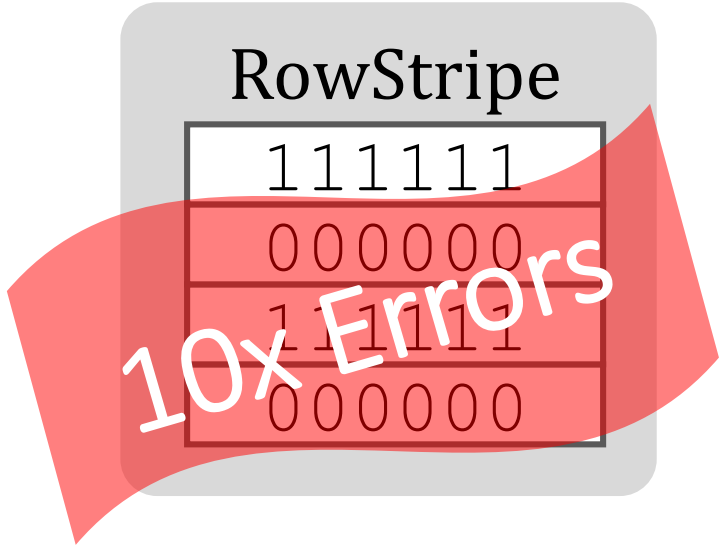
Slide credit to Onur Mutlu

Study RowHammer Characteristics

- Highly local nature of the bit-flipping capability

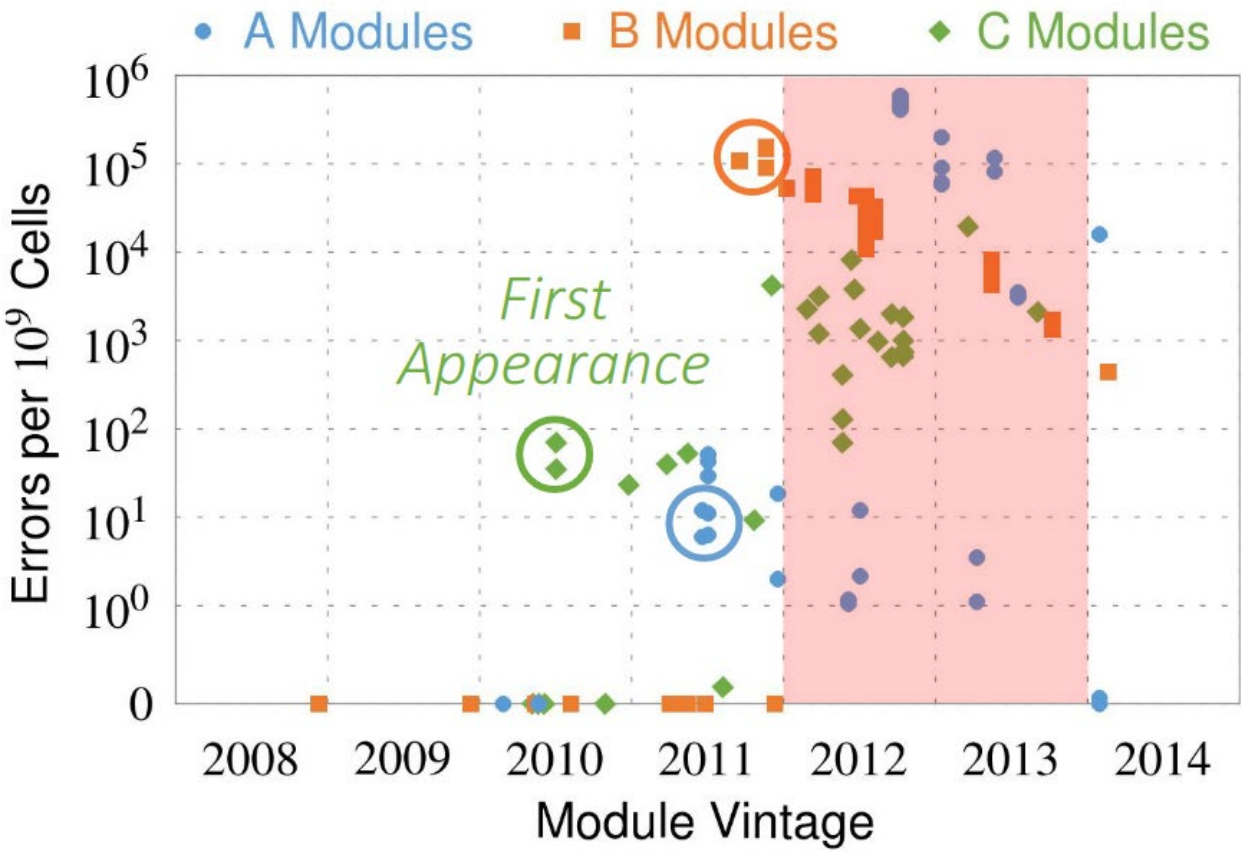
- The probability of bitflips are data-dependent





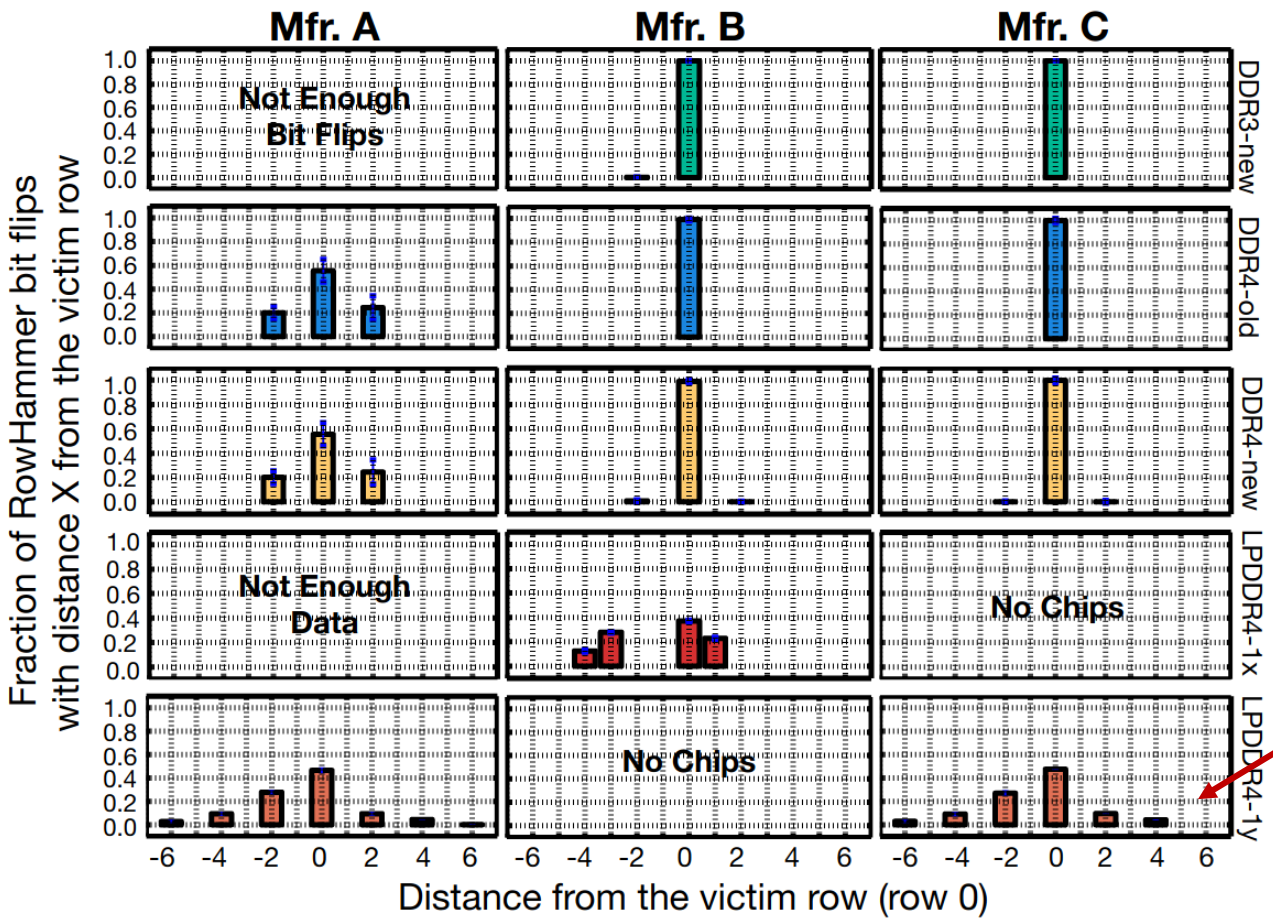
Slide credit to Onur Mutlu

Density Trends



- As DRAM gets physically denser, it becomes even **more vulnerable**!
- Only a few thousand hammer iterations are required on modern DRAM to cause a bit-flip

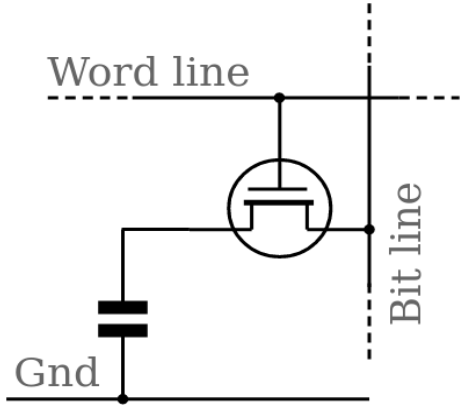
Density Trends

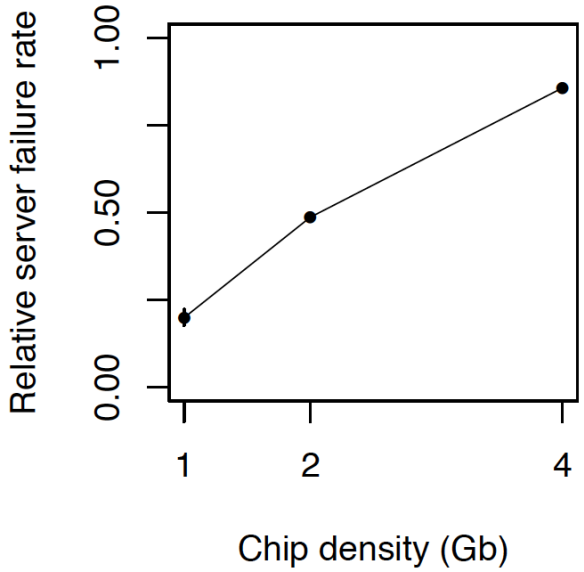


Denser DRAM also can result in flips in rows which are not *directly* adjacent to the attacker

Technology Scaling

- Capacitor must be large enough for reliable sensing
- The access transistor should be large enough for low leakage and high retention time
- Scaling beyond 40-35nm (2013) is challenging [ITRS, 2009]

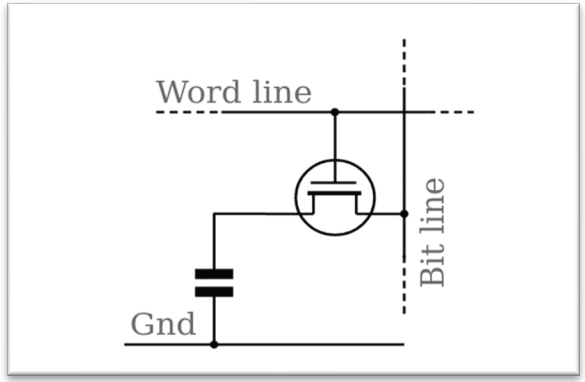




Data from all of Facebook's servers worldwide

Revisiting Memory Errors in Large-Scale Production Data Centers: Analysis and Modeling of New Trends from the Field; Meza. et al. DSN'15

Why Is RowHammer Happening?

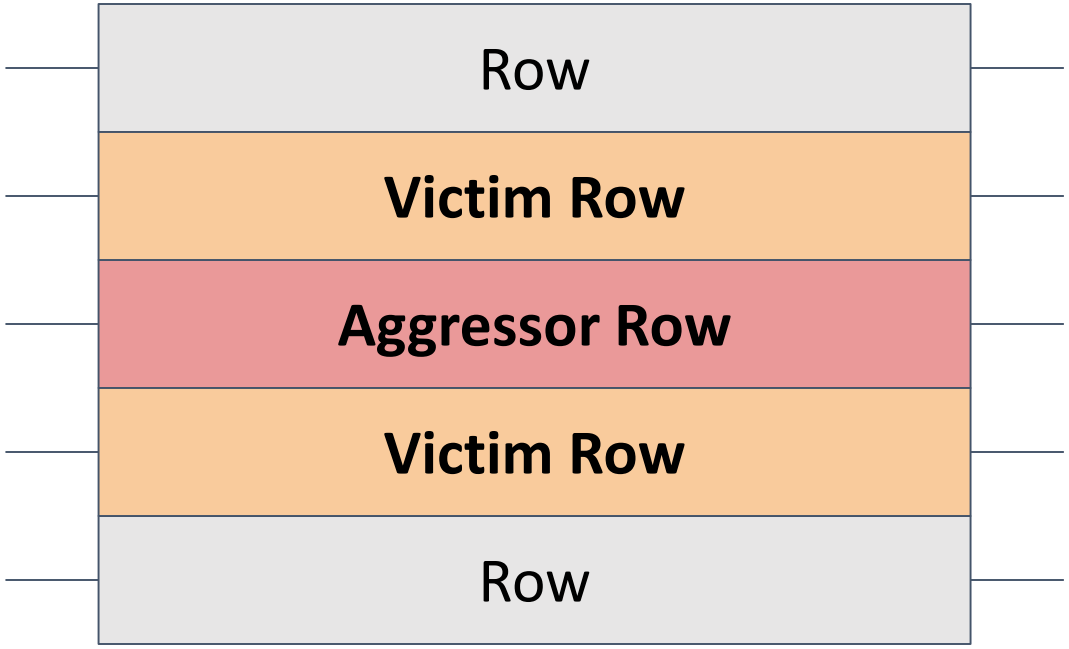


- DRAM cells are too close to each other
  - They are not electrically isolated from each other
- Access to one cell affects the value in nearby cells
  - Due to electrical interference between the cells and wires used for accessing the cells
  - Also called cell-to-cell coupling/interference
  - Other hypothesis exists
- Example: When we activate (apply high voltage) to a row, an adjacent row gets slightly activated as well
  - Vulnerable cells in that slightly-activated row lose a little bit of charge
  - If row hammer happens enough times, charge in such cells gets drained

RowHammer Attacks in Action

RowHammer Attacks in Practice

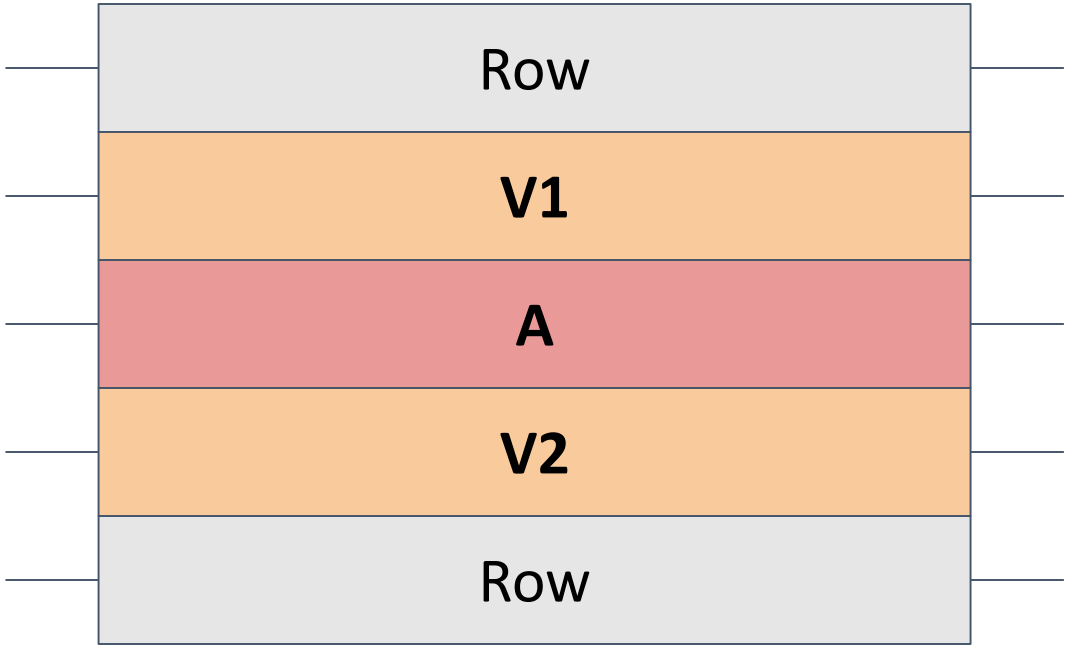
- Aggressor Row = Hammered Row



Challenges:

1. How to hammer? Need to access aggressor row enough times between refreshes.
2. Address mapping. How to find addresses map to neighboring rows?
3. How to make victim's data to be located inside vulnerable cells?

Hammer Attempt #1: repeat accesses

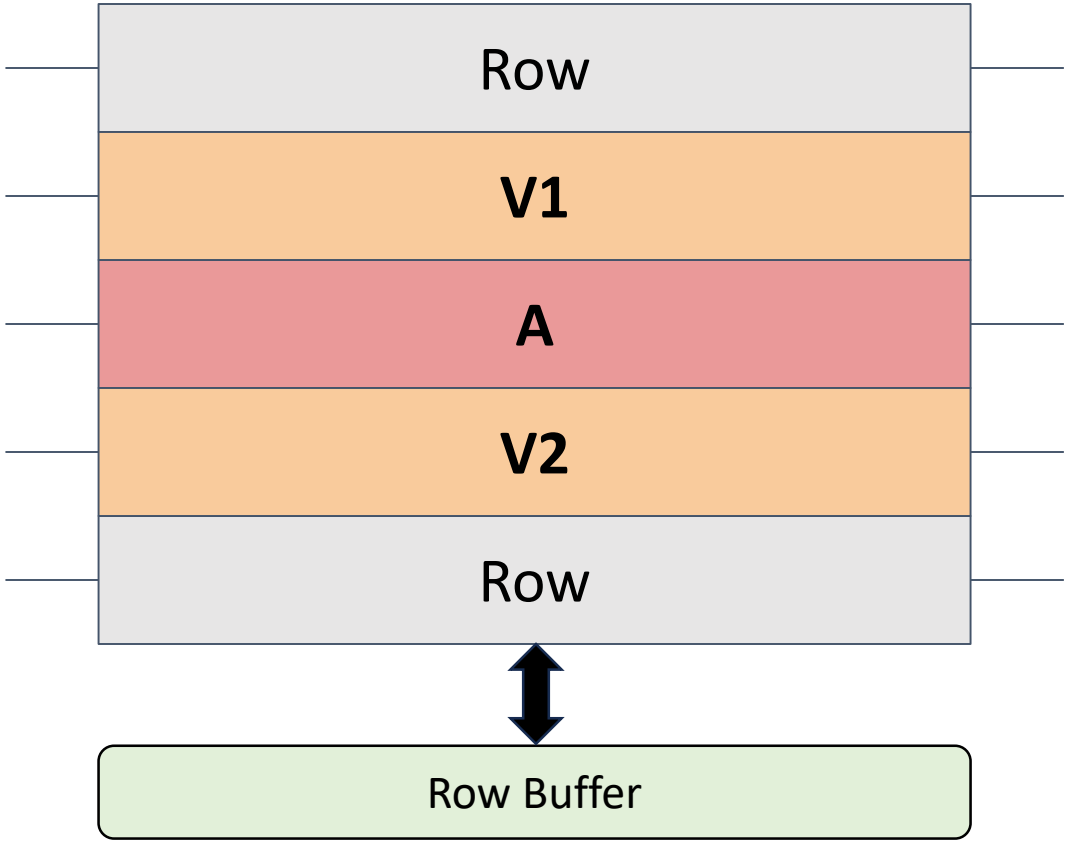



```
loop:
    mov (A), %eax

    mfence
    jmp loop
```

Will this work?
Why?

No. Because we will hit the cache.

Hammer Attempt #2: use clflush

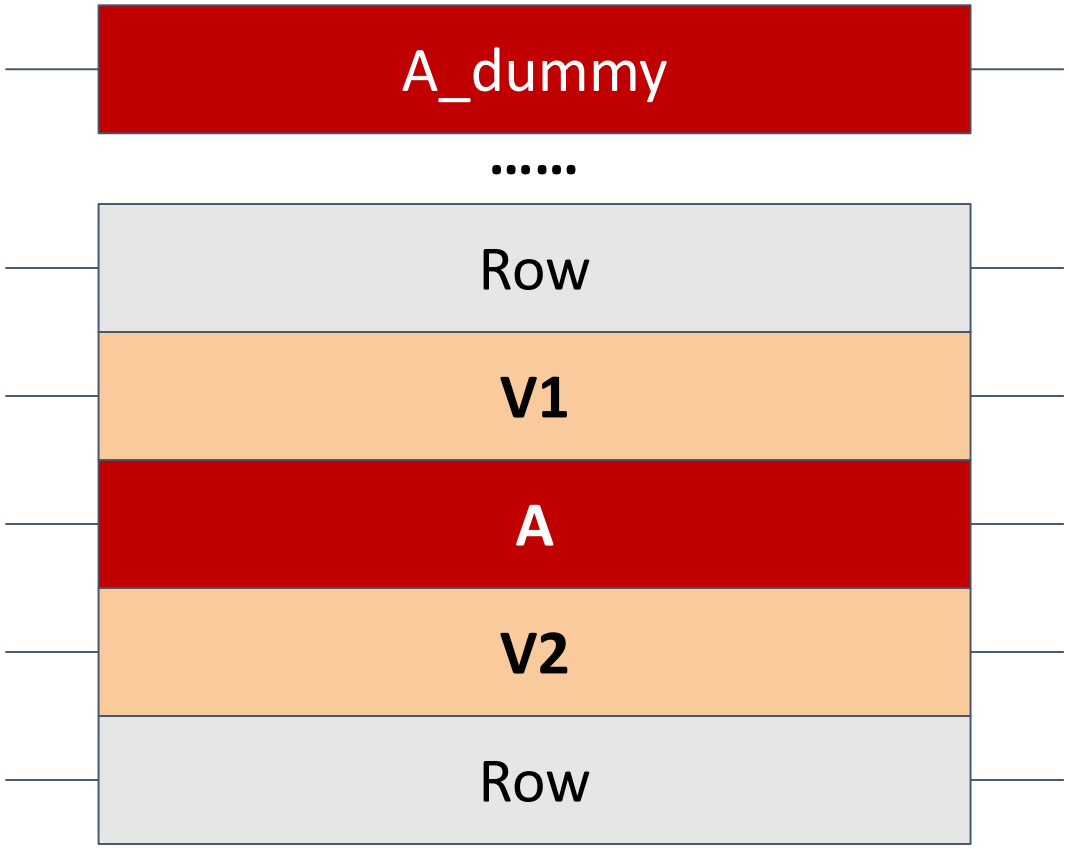



```
loop:
    mov (A), %eax

    clflush (A)

    mfence
    jmp loop
```

Will this work?
Why?

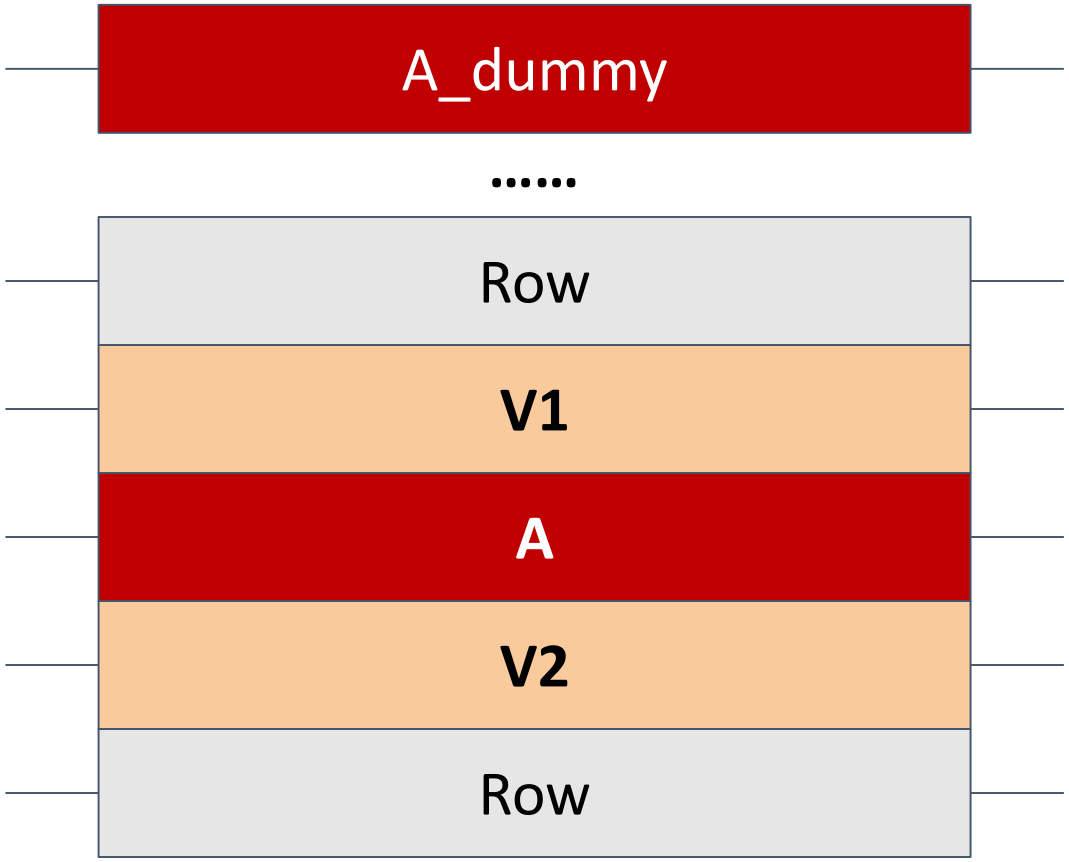
No. Because we will hit the row buffer.

Hammer Attempt #3: force row open/close




```
loop:
    mov (A), %eax
    mov (A_dummy), %ecx

    clflush (A)
    clflush (A_dummy)

    mfence
    jmp loop
```

“Single-Sided” Rowhammer

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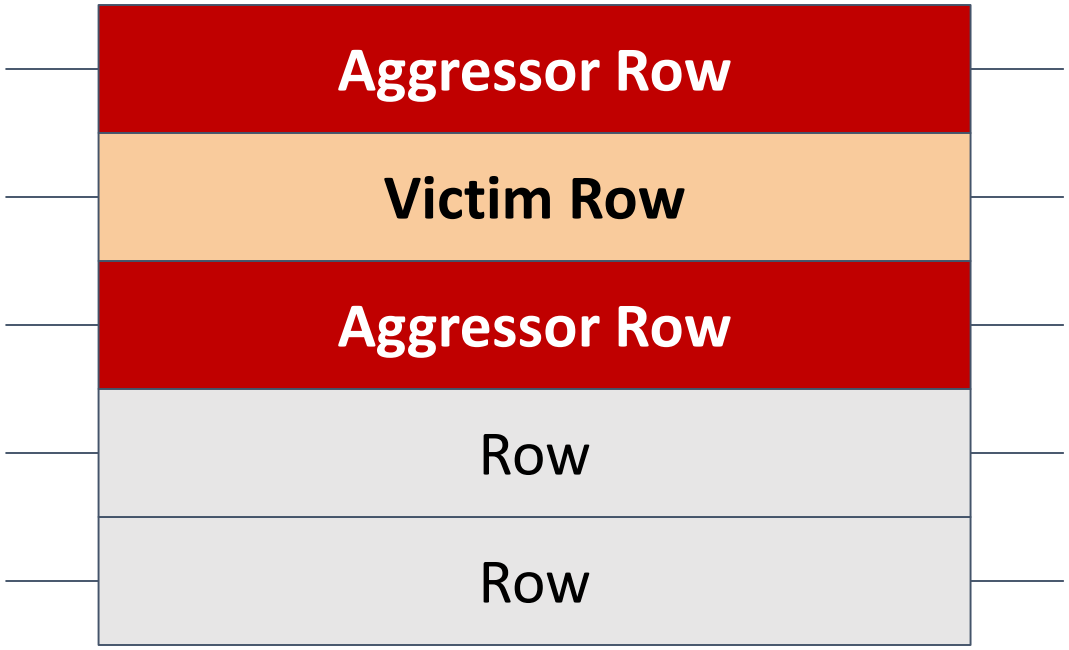
“Double-Sided” Rowhammer

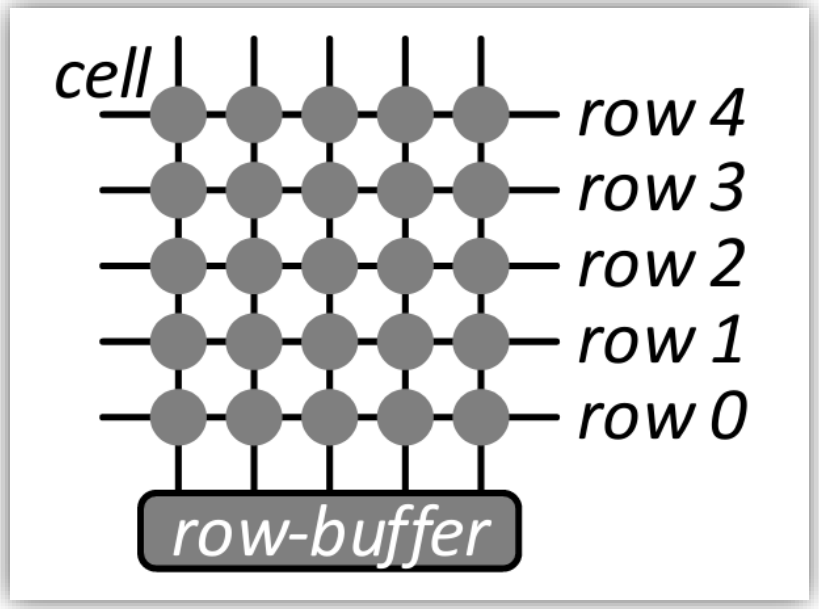


Increase the stress:

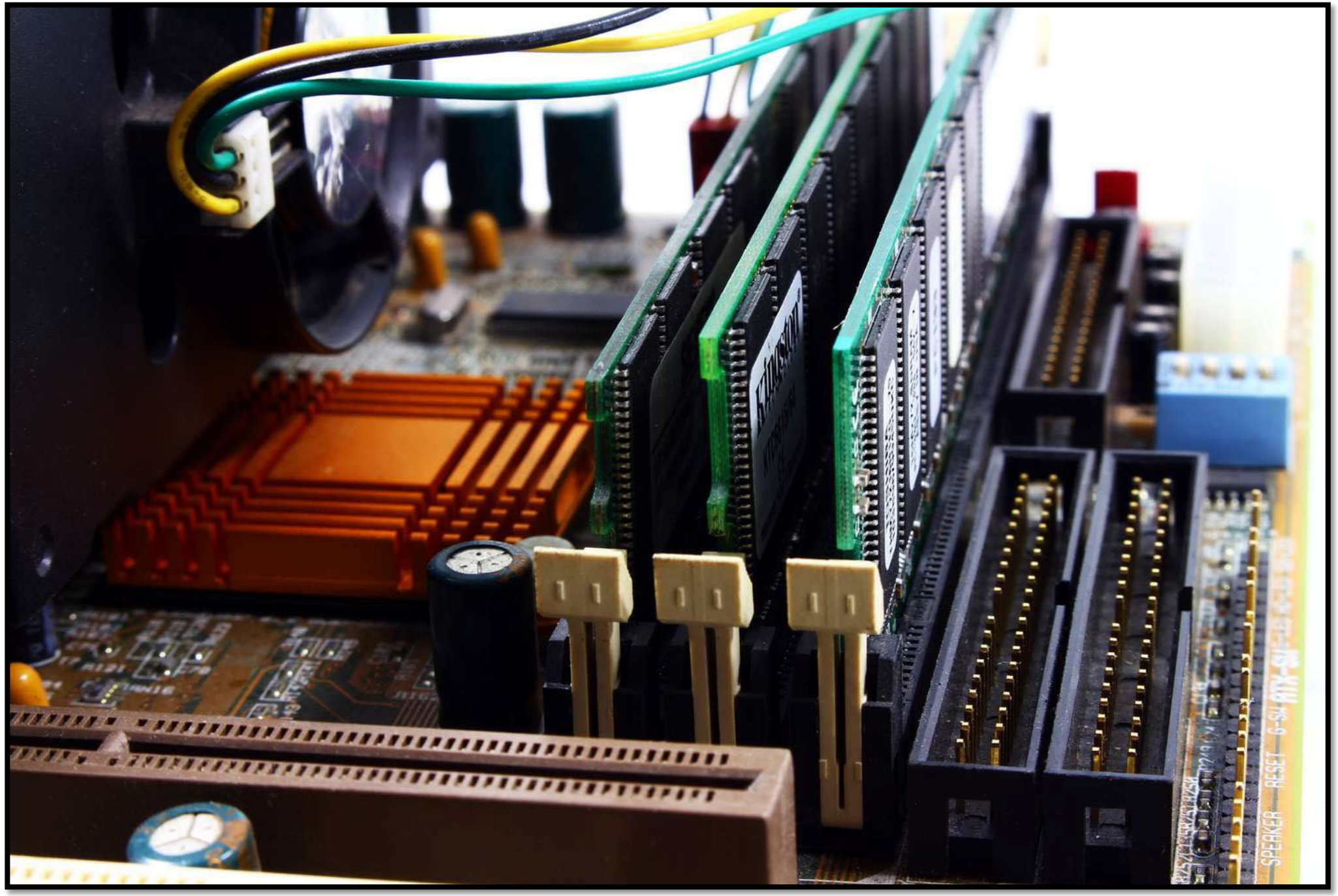
Repeatedly accessing both adjacent rows significantly increases the error rate of the victim row

Challenge #2: DRAM Addressing

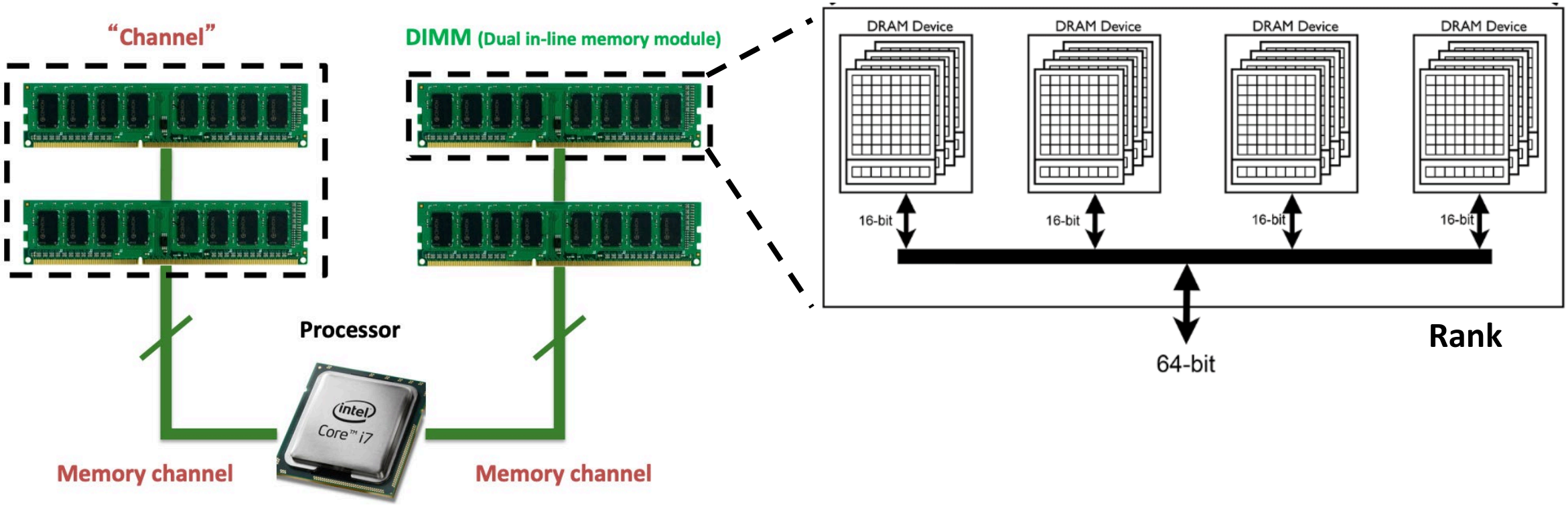
Virtual Address → Address translation Using Page table → **Physical Address** → ? Map to cells? →



Goal:
Find three addresses (2 aggressor addresses and 1 victim address) that map to consecutive rows in DRAM

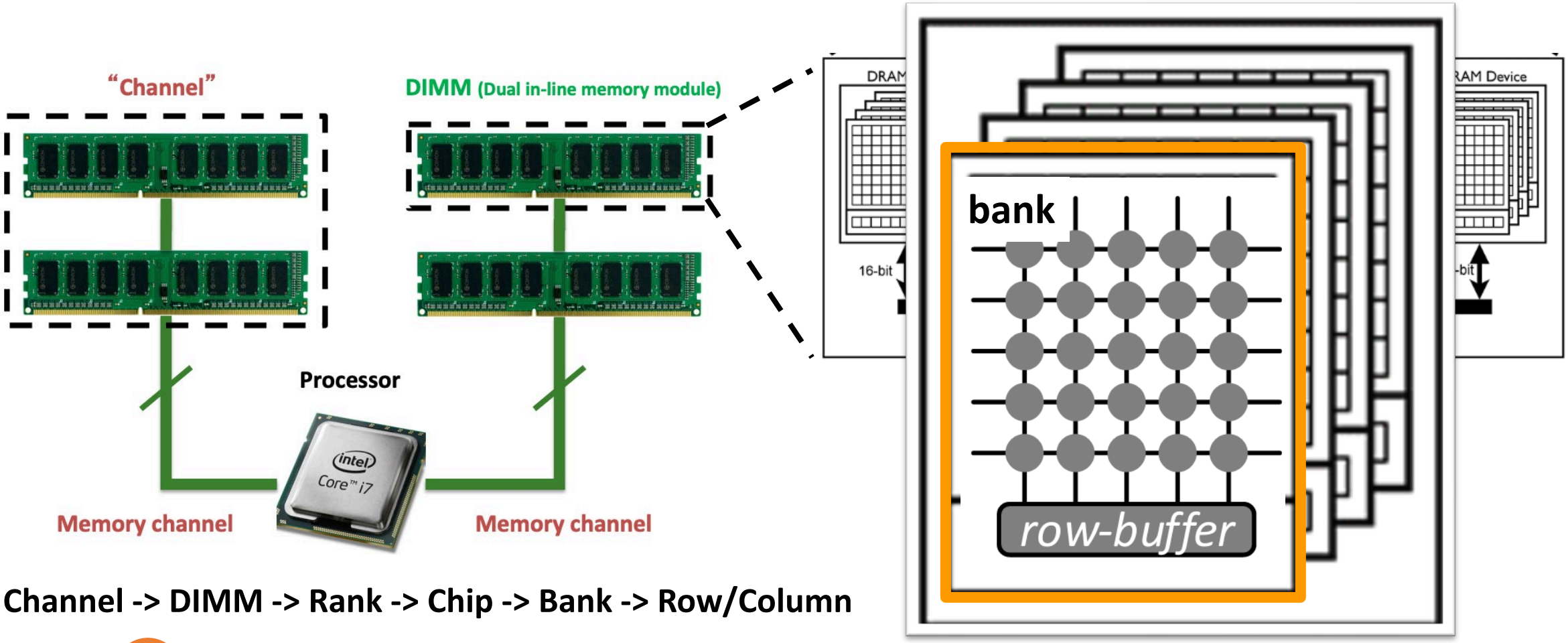

KINGSTON

DRAM Organization: Top-down View



Aggressor and victim addresses should have the same Channel ID and Rank ID

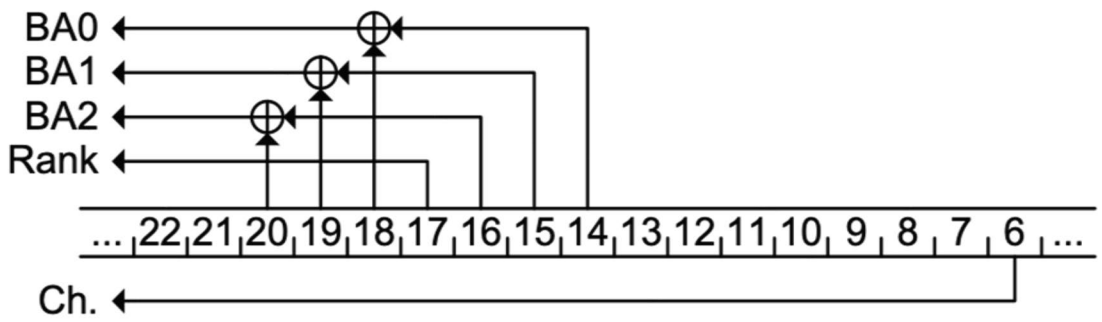
DRAM Organization: Top-down View



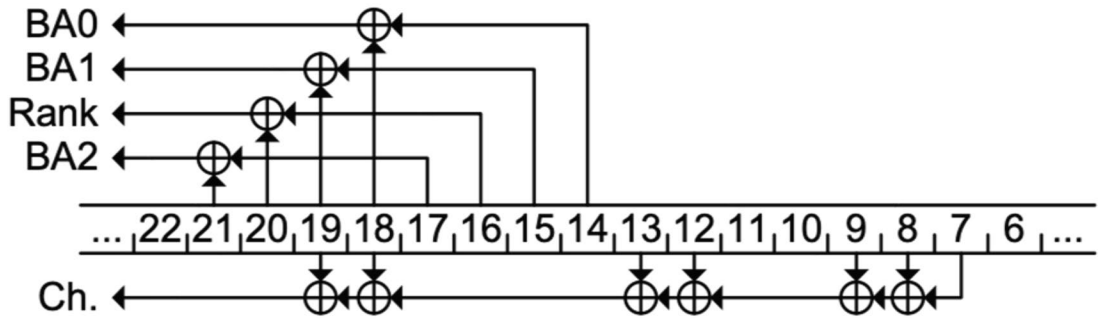
Channel -> DIMM -> Rank -> Chip -> Bank -> Row/Column

Aggressor and victim addresses should have the same Channel ID, Rank ID, and Bank ID

Address Mapping Examples



(a) Sandy Bridge – DDR3 [23].



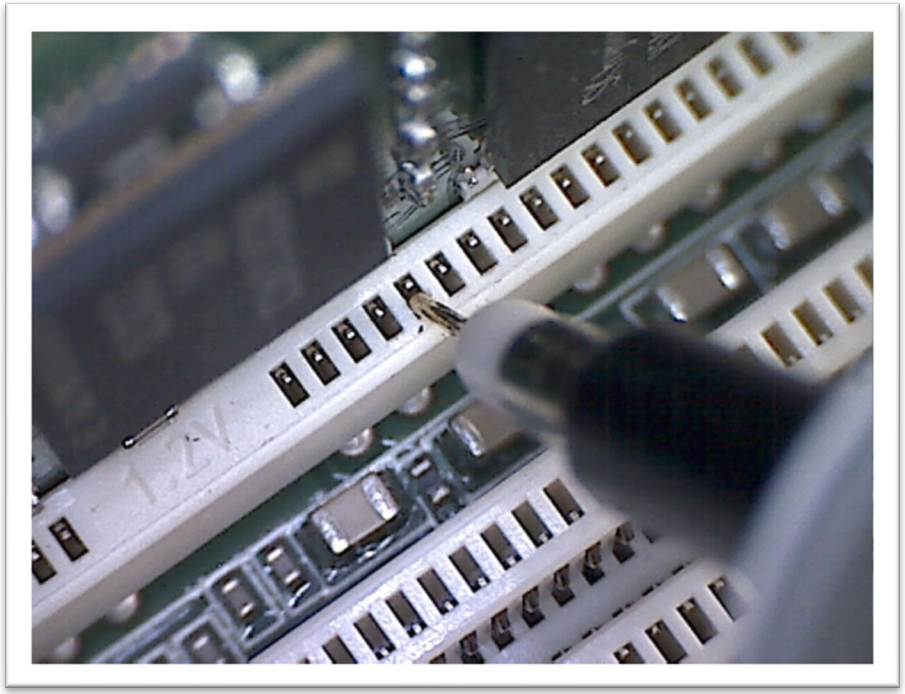
(b) Ivy Bridge / Haswell – DDR3.

Different machines use different mapping functions.
How to reverse engineer these functions?

Pessl et al. DRAMA: Exploiting DRAM Addressing for Cross-CPU Attacks. USENIX'16

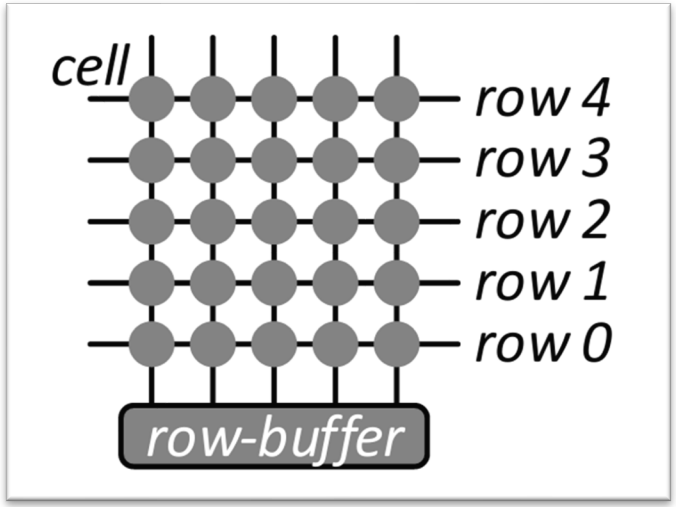
Reverse Engineer the Mapping

- Approach #1: Physical Probe

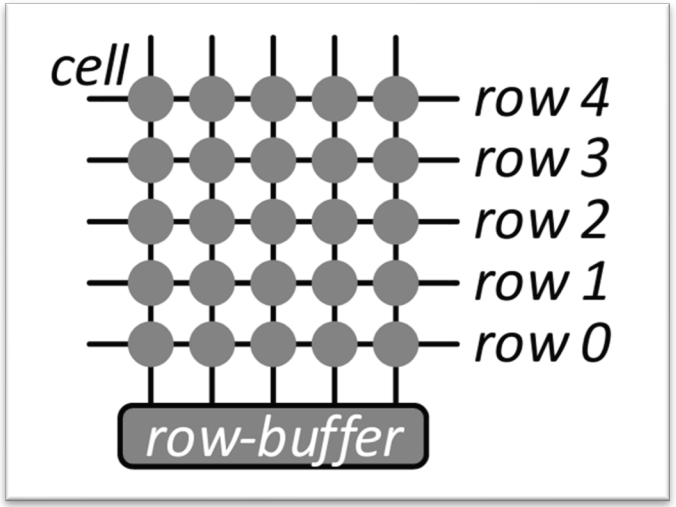


Reverse Engineer the Mapping

- Approach #2: Timing Side Channel via Row Buffer
  - Repeatedly access two addresses X and Y, and measure their access latency
  - What if both addresses map to Bank A?
  - What if address X maps to bank A and address Y maps bank B?



Bank A



Bank B

Attacks Built Upon RowHammer

Native Client (NaCl) Sandbox Escape

- NaCl is a sandbox for running native code (C/C++)
- Runs a “safe” subset of x86, statically verifying an executable
- Use bit flips to make an instruction sequence unsafe

Example “Safe” Code:

```
andl $~31, %eax   // Truncate address to 32 bits
                  // and mask to be 32-byte-aligned.
addq %r15, %rax   // Add %r15, the sandbox base address.
jmp *%rax         // Indirect jump.
```

Native Client (NaCl) Sandbox Escape

We can flip bits to allow for (unsafe) non 32-byte-aligned jumps

Exploited "Safe" Code:

```
andl $~31, %ecx   // Truncate address to 32 bits
                  // and mask to be 32-byte-aligned.
addq %r15, %rax   // Add %r15, the sandbox base address.
jmp *%rax         // Indirect jump.
```

Exploiting the DRAM rowhammer bug to gain kernel privileges (Seaborn and Dullien)

Kernel Privilege Escalation

What could happen if a user could gain direct write access to a page table?

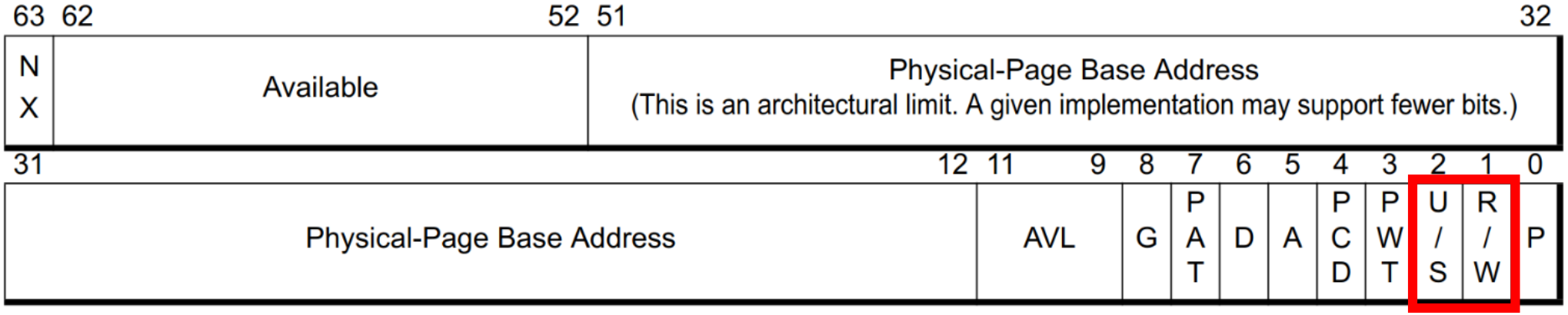


Figure 5-21. 4-Kbyte PTE—Long Mode

Steel SSH Keys

Observation: the likelyhood of Rowhammer bitflips can be influenced by the data pattern.

(b) Single-sided Rambleed. Here, the sampling page (A1) is neighbored by the secret-containing page (S) on a single side.

RAMBleed: Reading Bits in Memory Without Accessing Them; Kwong et al, S&P'20

Other Attacks

- Virtual machine takeover
  - Use page de-duplication to corrupt host machine
- OpenSSH attacks
  - Overwrite internal public key with attacker controlled one
- Drammer
  - Rowhammer privilege escalation on ARM
  - Utilizes determinism in page allocation to target vulnerable DRAM rows
- Rowhammer.js
  - Remote takeover of a server vulnerable to rowhammer

Without memory integrity, *any* software-based security mechanism is insecure!

Rowhammer Mitigations?

- Manufacturing “better” chips
- Increasing refresh rate
- Error Correcting Codes
- Targeted row refresh (TRR) - Used in DDR4!
- Retiring vulnerable cells
- Static binary analysis
- User/kernel space isolation in physical memory

Rowhammer Solutions?

- Manufacturing “better” chips — cost
- Increasing refresh rate — Performance, power
- Error Correcting Codes — cost, power
- Targeted row refresh (TRR) - Used in DDR4! — cost, power, complexity
- Retiring vulnerable cells — cost, power, complexity
- Static binary analysis — security
- User/kernel space isolation in physical memory

Takeaways

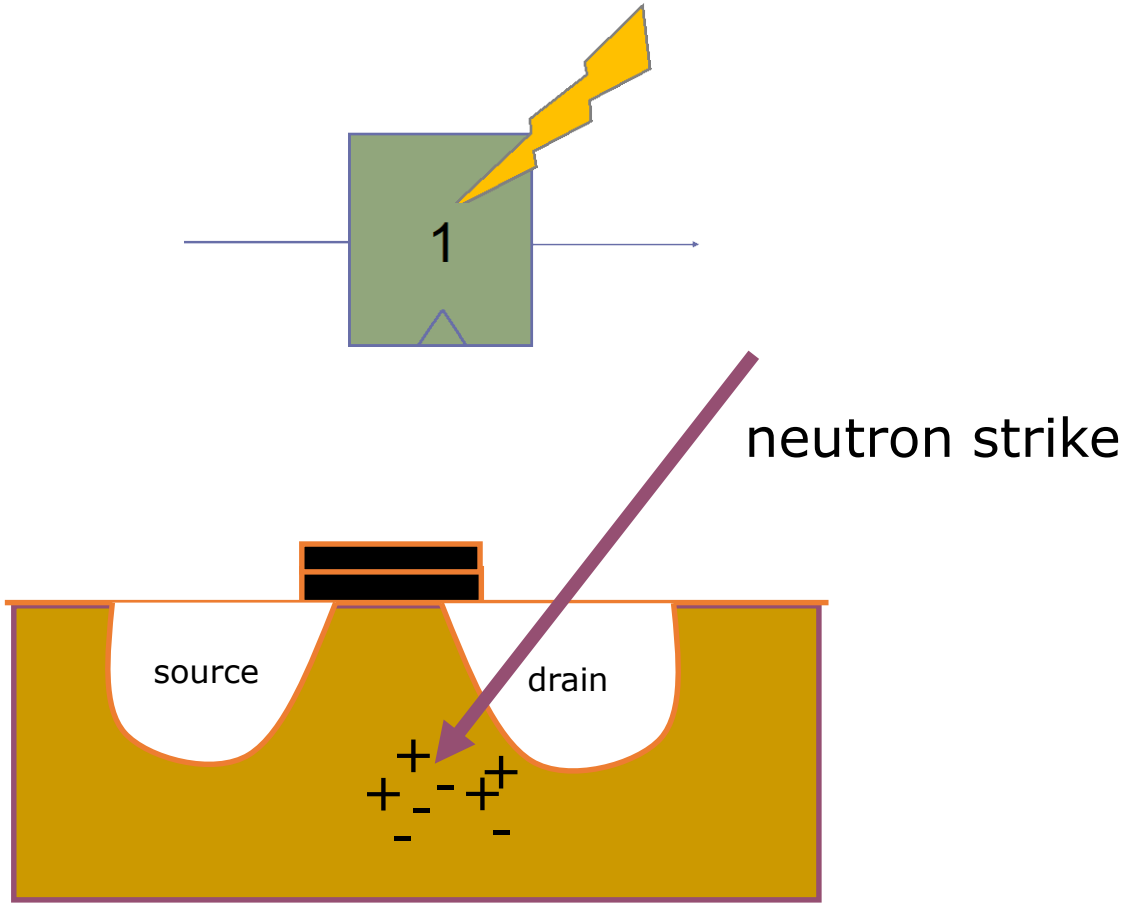
Reliability ←→ Security Implications

Stop Yield Speed Limit

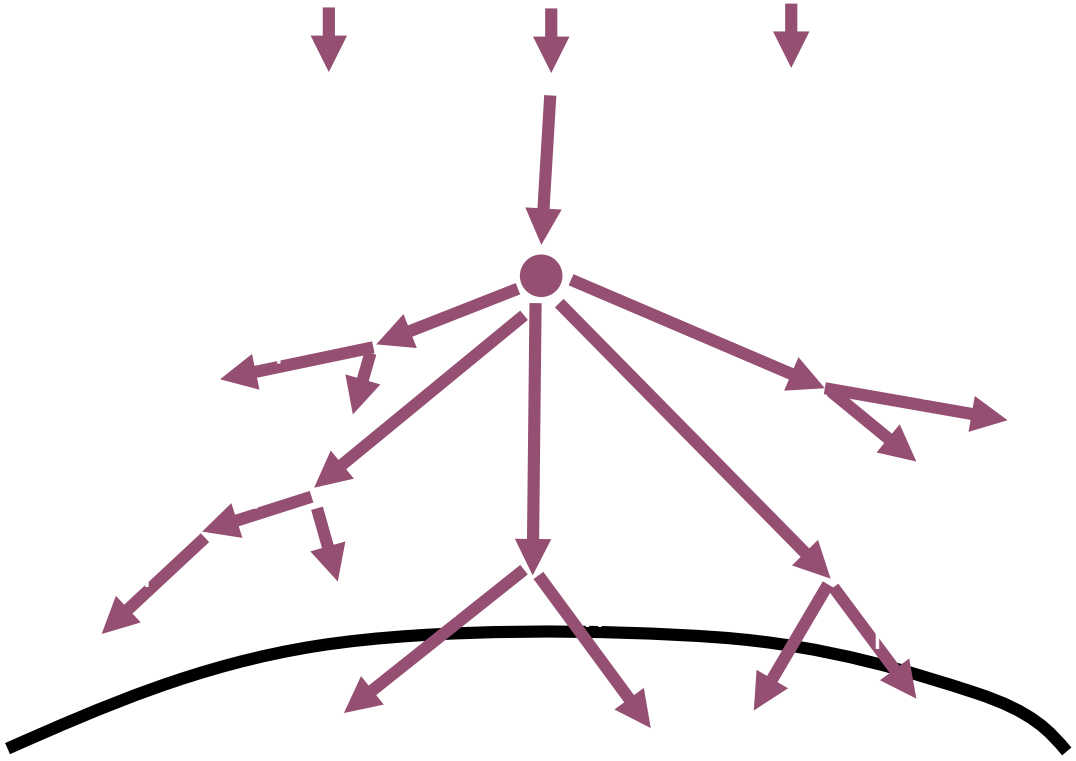
(a) Normal (b) Attack

Robust Physical-World Attacks on Deep Learning Visual Classification - Eykholt et al. CVPR'18

Reliability Problems in 2000s

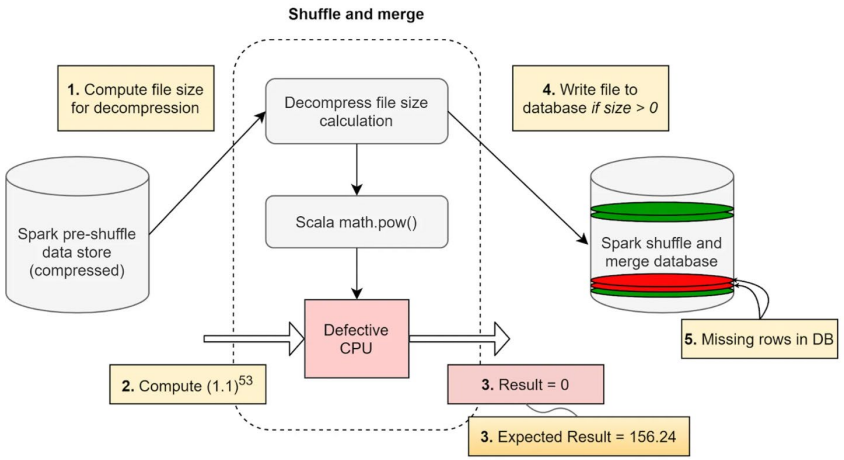


Transistor Device



Other Reliability Problems in 2020s

- Silent Data Corruption (SDC)
  - Cloud companies noticed SDC is a widespread problem for large-scale infrastructure systems.
- Problems
  - Long error detection latencies: taking days to weeks
  - Scalability



Example errors:

$Int\,[(1.1)^{3}] = 0$, $expected = 1$

$Int\,[(1.1)^{107}] = 32809$, $expected = 26854$

$Int\,[(1.1)^{-3}] = 1$, $expected = 0$

"Cores that don't count" by Google, HotOS, 2021
"Silent data corruption at Scale" by Facebook, Arxiv, 2021

Next:
Reliability Solutions